

Supplementary Material to: Universe24: Physical Relations from the Group Ring of a Cyclic Group under the Octahedral Group of Order 24

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This Supplementary Material accompanies the paper of the same title and is generated from the same source. The paper cites its sections as S1 to S6 and its tables as S1 to S3; the references are this document's own list; every other cross-reference (a theorem, a section, an equation, a table) is the paper's.

S1 The confrontation register, the full record

Table S1: The confrontation register, the full record, every row [31], with the law's own number and a status of three values, agrees, disagrees, not predicted (its missing piece and kind), defined in Section 7; the kinds in every cell: DETECTOR a count of clicks, the only measurement; GAMEBOARD a number of the board's state, a diagnostic; COMPUTATION the algebra's number, no run; the model's numbers are ideal, at the registered grain, not a prediction for a given instrument. The one count, repeated in the abstract and in Section 4: of the ten observables the law as built predicts (eleven once row 14 is read), four agree within tolerance (1a, 2b, 9; 5a as a bound met), six disagree (1c, 6, 7b, 8a, 8b, 13), and eight are not predicted by construction (3, 4a, 4b, 5b, 7a, 8c, 11a, 11c), each with its missing piece named in its cell and its kind, the instrument missing in the model, nature's number a bound or a fitted parameter, no click definition yet, or the model's own number a declared input; 2a NOT COMPARED until its source is verified; six rows keep their words (12 history, 2c not compared, 2a as said, 10 not yet, 11b a pin, 14 not yet read against nature), and 1b is a control, not counted; the hypotheses beside rows 6, 13 and 14 change no verdict of the law (the PASS rows replicated [6]). Each reading's kind is named: DETECTOR (a click), GAMEBOARD (the host's view, a diagnostic) or COMPUTATION (the algebra's); the criteria row by row, with the re-runs at head, are [28].

Row	Observable and nature's value	The law's reading, by kind	Status
1a	The CHSH sum of a pair: 2.42 ± 0.20 [48]; the quantum bound 2.828	$S = 2.75$ at $N_\phi = 64$, series L (DETECTOR; replayed at head by the test suite)	agrees: $S = 2.75$ at $N_\phi = 64$ (DETECTOR), 1.65 standard errors above the measured, 0.078 below the quantum bound; the pair by the click, Section 3 [30, section 10]

Row	Observable and nature's value	The law's reading, by kind	Status
1b	The same, the phase-form window	$S = 2$ exactly (DETECTOR, the counters' clicks)	a control, not counted: $S = 2$ exactly (DETECTOR), the local bound, 2.1 standard errors below the measured; Bell's bound is a theorem of every local read-out and the phase-form window is one, the law's answer to nature being row 1a disagrees: 15/16 and $-1/16$
1c	No signalling in the order: nature 0 on the second party's serial correlation at lag 1 under the first party's settings (the outcome order of a pair random, carrying no setting of the other party; the marginals' independence holds in nature and in the law alike)	15/16 under a constant $a = 0$ and $-1/16$ under the period-3 cycle $a = 0, 21, 42$, cyclic over 192 births on the registered pair world at $N_\phi = 64$ (DETECTOR, the run of 2026-09-22 [16]); the marginals 96/96 under both (DETECTOR); the algebra's 15/16 and $-1/16$ met at 16 of 16 checks; the cycle $N_\phi/2, 0, 0$ of Section 4.5, $-5/16$, a computation the register's keys cannot declare	(DETECTOR) against nature's 0; the law's determinism, the wheel a counter, shows in the order and not in the counts
2a	Two-slit visibility of one quantum at a time: 0.98, the register's source [49] a Mach-Zehnder reading and not a two-slit one; a two-slit source, a Fresnel biprism's central fringe at 0.94 as reported [29], its figure to be verified against the source; the ideal 1; the criterion the clicks' visibility $(I_{\max} - I_{\min})/(I_{\max} + I_{\min})$, no apparatus model	0.966 in the clicks of <code>slits_huygens</code> (the birth wheel, 4096 births; L2b; DETECTOR), the pin derived for this world's fan by the register's own algebra 0.9659 (COMPUTATION) and met bit for bit by the re-run at head, 0 of 121 pixels differing; the record's weights read the same under the exact phase (the 0.954 of the built-phase pin of 2026-09-20 is history)	NOT COMPARED until the two-slit source is verified; the law's number 0.966 in the clicks (DETECTOR) a prediction under every fan (the grain's share at most 12 percent of the shortfall, 0.004 of 0.034, the kind audit [11, section 3], COMPUTATION); a bound met against the biprism's 0.94 at the abstract's level [29]; against the Mach-Zehnder's 0.98 another geometry
2b	Mach-Zehnder visibility: 0.98 [49]; the criterion the visibility $(b - d)/(b + d)$ of the bright and dark ports' offers of <code>mz_equal</code> , no apparatus model	the offers 1681/1682 and 1/1682 (GAMEBOARD, the record's ledger); the clicks 64/0 over 64 births (DETECTOR; replayed at head by the tests)	agrees, a bound met and not a quantitative agreement: the clicks' visibility 1.000 (64/0, DETECTOR) above the apparatus-limited 0.98; the offers' 0.9988 a GAMEBOARD diagnostic of the apparatus layer
2c	The power of the click's form: the Sorkin parameter $\kappa_S = 0.0064 \pm 0.0119$ [37]	the power's window [1.917, 2.012] from the click cells (DETECTOR), a read-back of the built click (COMPUTATION)	NOT COMPARED (κ_S not computed): the window an implementation gate, no evidence about nature; no three-opening world is registered

Row	Observable and nature's value	The law's reading, by kind	Status
3	The deceleration parameter $q_0 = -0.53 \pm 0.01$ [38, 39, 40]	−0.108 coasting (series G2, the pointer's z : read at head in the detector's own clock, its count of self-creations under the age wall ($r = 1.0000$, no crowd at the detector; [28], the re-run of 2026-09-22), equal to the host's interval to the integer, DETECTOR; the registered run's time base was the host's interval [4, record 768], superseded; a two-way light clock at the detector, not read, reads the same count in no crowd (series X)); −0.104 at head ([4, record 408]; the re-run at head, the pin met, DETECTOR), the reading's own grain 0.004 between engines and 0.03 between windows	not predicted, kinds (1) and (2): the law's number −0.108 coasting (DETECTOR, the detector's own clock), the coasting ideal $q = 0$ within the reading's grain, 0.42 from nature's value; the crowd's history is missing, no term carries q toward −0.5 and the law's own crowd gives the wrong sign; nature's -0.53 ± 0.01 is a fitted parameter of the diagram
4a	The muon's lifetime in flight over its lifetime at rest: $\gamma = 29.33$ at the storage ring, the dilation confirmed as γ to a fractional error of about 2×10^{-3} [33]	the become at the count 64 at every speed (a body's own record, DETECTOR; its interval GAMEBOARD), the products' face clicks (DETECTOR): the ratio 1; under covariant-readings-v1 within its domain, the products' clicks 367 and 344, the ratios 70/64 and 124/64 against $\gamma = 1.1074$ and 1.9558 (DETECTOR) [8]	not predicted, kind (1): no moving thing's clock rate, $r = 1$; the law's number 1 against 29.33; beside it, as a declared hypothesis, γ 's form within the domain $\gamma \leq 2$ [8], nature's 29.33 outside the domain under either drive, a factor 14.8 short at the cap
4b	A moving lamp's redshift with the clock's factor [34]	$z = 0.2636$ at $\beta = v/c = 0.2674$ (series G2: read at head in the detector's own clock, its count of self-creations under the age wall ($r = 1.0000$, no crowd at the detector; [28], the re-run of 2026-09-22), equal to the host's interval to the integer, DETECTOR; the registered run's time base was the host's interval [4, record 768], superseded; a two-way light clock at the detector, not read, reads the same count in no crowd (series X)); 0.2647 at head (the re-run, the pin met, DETECTOR)	not predicted, kind (1): the law's number 0.2647 (DETECTOR, the detector's own clock) against 0.315, 17 grains below, $1 + z = 1 + v$ with no moving clock's rate ($r = 1$); beside it, under covariant-readings-v1 as a declared hypothesis, 0.3674 for 0.369 ± 0.003 , read at a click over the detector's own count, 0.3674 in [0.366, 0.372] [8]; the two one-way factors apart by $1 - v^2$ under the law and at 1 within 0.4 percent under the key, read on the k -ladder [8]
5a	The anisotropy of c by direction, below 10^{-18} [32]	the flight table's pace 0.5774 to 0.5818 at $N_l = 64$ (COMPUTATION; series Q's 290 of 290 clicks within it at finite ages, DETECTOR, Appendix B)	agrees, a bound met: the pace 0.5774 to 0.5818 by direction (COMPUTATION; series Q's 290 of 290 clicks within it, DETECTOR), isotropic within $1/T_D$; a bound on N_l of order 5.8×10^{17} for nature's 10^{-18}
5b	The anisotropy of c between a laboratory's two arms when the laboratory moves: the null at the Earth's $\beta = 10^{-4}$ [47]	the round trips γ^2 along the motion and γ across on the flight table, 1.375 and 1.140 at $\beta = 0.4297$ (COMPUTATION, a pin; the pair in motion not run)	not predicted, kind (1): nothing contracts, the bond a whole Link and a contraction a root of the state at run time; the pin's arms differ by $\beta^2/2 = 5 \times 10^{-9}$ at $\beta = 10^{-4}$, eight to nine orders above the bound

Row	Observable and nature's value	The law's reading, by kind	Status
6	Bohr's line ratio $\nu(\text{H}\beta)/\nu(\text{H}\alpha)$ of the Balmer series: 1.3500, 656.279 over 486.135 nm [27], Balmer's 27/20 exactly	by algebra first, on the law's integers (COMPUTATION [22]): a closed loop is an integer congruence of the action row, $\Delta A_a = 0 \bmod h$ on each axis with the quotients summing to jN ; two closing radii stand as the square of two counts a detector reads, $a_j/a_i = (j/i)^2$ in the shell mean, checked by $T_j/T_i = (j/i)^3$, no root, no float; from a closed circle one click reaches only the loops with $2i^2 \geq j^2$ and a release of content unbinds, so no line of Balmer's is one click of the law: NEEDS A NEW RULE. The law as built: the electron escapes at 3407 intervals (series H's world, DETECTOR; the cause the half-Link lag [18]). Under the key centred.step (a declared start of the drive, off by default; Section 3): the loop stays, no escape face click in 7500 intervals, 23 quarter crossings (C1 PASS, DETECTOR), tightening from 12 to 7 Links at two crossings over five turns, the period falling from 1431 to 1040 (C2, C3 FAIL as declared, DETECTOR; the run of 2026-09-22 [17]); no line read	disagrees: the loop opens at 3407 intervals (DETECTOR) and no line is read, against Balmer's 27/20; under the key C1 PASS, C2 and C3 FAIL, no pin moved; no rule of the six quantises a body's loop, so the lines are computable by no run of the law as built and 27/20 waits on a rule under its own identity; the ladder $a_j/a_i = (j/i)^2$, $E_j \propto 1/j^2$ and Balmer's 27/20 conjectured from the algebra in the shell mean's limit, not read on the board; a lattice able to hold the scale (the orbit's radius sixty thousand times the nucleus's) is not run (Section 7.2 (viii))
7a	The deuteron's binding fraction 0.1185 percent [35]	the bond's two border clicks and the mass read 3673 (DETECTOR) against the declared 3677, 4 units, 0.109 percent (series N); the books' escaped 4 (GAMEBOARD, a diagnostic) the check; the same at head, every pin met	not predicted: the give per nucleon a declared input read back [4, record 106]; the escaped 4 of 3677 (DETECTOR) brackets nature's 4.35 between the whole gives 4 and 5; the missing piece a rule that derives the binding; the law's number the mass read 3673 against the declared 3677 (DETECTOR), 0.109 percent
7b	The alpha's binding over the deuteron's, 12.72 [35]	2.0 (series N, the border's four clicks against two, DETECTOR; the same at head)	disagrees: 2.0 (DETECTOR) against 12.72, a factor 6.4; the give once per body
8a	The neutron's decay curve, exponential, width over median 3.17 [45]	a step: every neutron at its key on its own clock, the width 0 (DETECTOR, the 64 clicks of content 3); the width over the median in the lattice's clock 0.036 in the registered run and 0.08 to 0.13 at head at the cap 1024 (GAMEBOARD, the lattice's clock, counted in no criterion; against nature's 3.17 these widths would read a factor 88 in the registered run and 25 to 40 at head, numbers of the diagnostic that stand in no verdict)	disagrees: a step, the width 0 (DETECTOR) against nature's 3.17, the step the law's own prediction; a width would need a decay fired by a met row of a declared bath ([7, section 1.8]), none of the three pieces; the width in the lattice's clock a diagnostic beside it; the registered J1 run predates the one flight wall, under which the weak register's become blocks move ([4, records 931 to 933]), J1's redeclaration in the weak field after the paper

Row	Observable and nature's value	The law's reading, by kind	Status
8b	The neutrino's passage through a second detector, about 1	16 of 1024 with 0 behind (series J2, a window; DETECTOR; the same at head, the pin met bit for bit)	disagrees: 16 of 1024 and 0 behind (DETECTOR, series J2) against nature's about 1; under the declaration massive-rows-v1 with the ν quantum 1 the readers read 0 and 0 and the reader at the Node of a whole turn the whole beam, 1013 of 1013, the far detector 0, by the law's design of the path phase, the beat $h/p = 4.65$ Links read at a click for the first time [9]; beside it, with a re-emitter declared between the source and the readers, an arrangement of the law and not nature's, the second reader's rate over the first's is 1.000 exactly (16 of 1024 and 16 of 1007, the far detector 683 of 705, DETECTOR) [9], not a pass against nature
8c	The heaviest neutrino mass state over the electron's mass: at least 9.8×10^{-8} [42, 43]	the ν family with no content, an input	not predicted, kind (2): the family's content is a declared input, and nature's number a bound; the law's number 0, the ν family with no content, refuted as an input by the observed splittings; under massive-rows-v1 the quantum 1 is the input no longer 0, a computation and no click, the electron's declared content then at least 1.02×10^7 units (COMPUTATION) [9]
9	Malus's law at 45 degrees, 1/2	128 of 256 exactly; 219 of 256 at 22.5 degrees (A12; DETECTOR, replayed at head by the tests)	agrees: 128 of 256 exactly at 45 degrees (DETECTOR), exact at 45 and 90; 0.8555 against $\cos^2 = 0.8536$ at 22.5, within the tables' grain
10	The single-opening spread $w \Delta(\sin \theta)/\lambda$: 0.886, the full width at half maximum of the single-slit pattern [46]	the register's entry A10: 1.08 at $w = 27$ with $\lambda = 4.619$ Links (the crowd form, kept as history); the record form's re-run did not complete	NOT YET under the one click, as it is: the last registered 1.08 against 0.886, 22 percent above; the smaller widths unread
11a	The deceleration parameter read from the brightness of a far lamp: -0.53 ± 0.01 (row 3's sources)	a derivation's pin, $q_{\text{eff}} = +1$: the click count falls as $1/(1+z)$ and the content per click stays the birth's, one factor where nature's flux has two (COMPUTATION; the run not made)	not predicted, kinds (1) and (3): no term of the crowd's history, and no click definition of a brightness yet; the pin $+1$ against -0.53
11b	The stretch of a far lamp's stream over $1+z$: the exponent $b = 0.97 \pm 0.10$ of $(1+z)^b$ [41]	the same pin: 400 rows released over 400 intervals arrive over 1452 at 300 Links, the factor 3.639 against $e^{Hd/c_0} = 3.629$, $b = 1$ (COMPUTATION; the run not made)	PASS by its pin, as it is: $b = 1$ within 0.97 ± 0.10 ; pinned, the run not made, a pin needing a hypothesis not built
11c	Tolman's test, the surface brightness of a resolved source at z : $(1+z)^{-4}$, measured as $(1+z)^{-n}$ with n between 2.3 and 3.1 before correction [44]	the same pin: the board's Nodes and the fan's lines fixed under the wall, a ruler of l Nodes at d subtends l/d , the flux one power of $1+z$, $n = 1$ (COMPUTATION; the run not made)	not predicted, kinds (1) and (2): an angular size growing with the redshift is a board whose Links stretch, no verb on the state; nature's exponent a fitted value; the pin $n = 1$ against 4, three powers of $1+z$ at most

Row	Observable and nature's value	The law's reading, by kind	Status
12	The clock's field at two distances at the same push: the potential's form, 2.00 (the GPS term 45.7 microseconds per day [12]; [13, 36])	series T: the form measured at a detector at two Nodes (the ratio of the two records' rates over the same intervals, in which the interval cancels (DETECTOR)), the age clock 1.907 (the lines' pin 1.909 ± 0.05), the presence clock 1.000	read under the law before the generic entry of 2026-09-22 [19]; not readable under the law as it stands; the four worlds are being redeclared in the weak field (the pair [1, 16384], the crowd's age moment at most 1024 per Node on the light's path) and re-read, the new numbers to replace these; HISTORY, not a pass: under the age word (P9) as the form at two Nodes the lattice pin 1.909 ± 0.05 is met (DETECTOR, a gate on the code, the law before the entry), the lattice's dwelling ages 5, 6 and 10, 11 pinning 1.909 for the continuum's 2.00; against nature the reading 1.907 is 4.6 percent off with no uncertainty stated on either side, so the caption's PASS criterion is met nowhere, PENDING the weak-field re-read; FAIL under the presence word

Row	Observable and nature's value	The law's reading, by kind	Status
13	The bending of light beside a mass, the full deflection over the time part's alone, $1 + \gamma = 2$ at the Sun's limb, 1.75 arcseconds against the time part's 0.87 [26]	the law as built: a row reads nothing of the crowd, the deflection 0.000 pixel and the delay 0.00 interval (series K, three worlds, DETECTOR). By algebra first, on the law's integers (COMPUTATION): the walk of a light row past a held mass under the key optical, read at a ring of Nodes at the impact distance b, recovers the M/b form with $C_{\text{ring}} = \alpha bc^2/(GM) = 5.12$ at $b = 6$ (5.44, 5.86 at $b = 3, 8$), not 4 within the ring's grain, the factor left the fan's $F_{L1} = 1.421$ of the flow [20]; under the one constant (flow-link-v1, the flow counted per Euclidean Link of its line) $C_{\text{ring}} = 3.65, 3.82, 3.92$ at $b = 6, 3, 8$ against 4, that is $2c_f$ within the finite path's $L/\sqrt{L^2 + b^2}$ (0.974 at $b = 6$) and the shells' 0.990, one to three grains, $4 = 2(1 + \gamma)$ with γ an input [21]. The run of 2026-09-22 on the split ladder (the wall's square reached without the root; the first run's deciding ring had been refused by the working bound) [23]: Under flow-link-v1 (the key flow.link, off by default) the six ring worlds read their mean radial shifts at the algebra's numbers ($b = 6, \gamma = 1$, the deciding pin: 0.731 Links against 0.731 ± 0.025; every arrival Node the map's, 168 of 168; DETECTOR), and the constant through the stated lever-arm factor, $C_{\text{ring}} = 3.655, 3.819, 3.918$ at $b = 6, 3, 8$ and $\gamma = 1$ (COMPUTATION), is the algebra's own walk to 0.01 and 2.40, 1.07, 1.40 grains from the continuum's $2c_f \times 0.990 \times L/\sqrt{L^2 + b^2}$, outside the one-grain pin: the engine walks the rule as the algebra walks it, and the bending's 4 does not close within the grain (3.655, 3.819, 3.918 bare against 4); $c_f = 2$ is an input, $nS = d$ a declaration, 4 on the comparison side; nothing here is compared with nature.	disagrees: the law as built 0.000 pixel and 0.00 interval (DETECTOR, series K) against 1.75 arcseconds; beside it, under the declaration $\gamma_{\text{PPN}} = 1$ (the weight rule, stated once in the limitations paragraph), the ring's rule read inside its pins, the deciding ring 0.731 against 0.731 ± 0.025 (DETECTOR), the coefficient $2(1 + \gamma_{\text{PPN}})$ a declared input equal to nature's, read within the fan's comb (3.65, 3.82, 3.92 against 4, two to nine percent) and never within the grain, a declared input and not a prediction; the ring's shift and the ratio read the declared γ_{PPN} back (DETECTOR readings of a declared input); the coefficient on the one constant stays conjectured from the algebra

Row	Observable and nature's value	The law's reading, by kind	Status
14	Newton after a detector: the ratio of the periods of two closed loops at $r = 24$ and 12 under the $1/r$ push on the plane, 2.00 (the form's scale symmetry; nature's inverse square in space $2^{3/2}$); the register's bracket 2.00 ± 0.18 ; the run's loops never closed, so its ratio is not compared with nature	by formula first (COMPUTATION [21]): under the one constant every push is divided by the plane fan's mean of $S_1/ \mathbf{D} $, $F_{\text{plane}} = 1.2871$, so the circular balance solves to the whole $n = 8$ and the periods $T = 2\pi r(S + n)/n$ move from 343, 687 to 384, 768 (377, 754 at the whole 8), the ratio 2.00 unchanged, the factor a constant of the line cancelling in it. The run of 2026-09-22 under the key <code>flow.link</code> , series D3's four worlds, the pins by the generator before any run [24]: the controls inside (every click at $x = c_x + r$; the escape tick 305 in 299 to 311; DETECTOR); the probes escaped, <code>r12.flow</code> through the face $+y$ at the tick 1208 and <code>r24.flow</code> through the face $-x$ at 2452 (face clicks, DETECTOR, read from the registered events by the diagnosis [25]; the run's first record had read none), so no loop closed, and the readings called periods are the means of the recurrences of x before the escape, two and three unequal spacings, outside: $T(12) = 588.7$ against 343.1 to 410.9, $T(24) = 987.3$ against 686.1 to 821.8, $\omega^2 = 7.775 \times 10^{-5}$ (the angular rate squared) and 3.112×10^{-5} against 2.338 to 3.354×10^{-4} and 5.845 to 8.386×10^{-5} , the amplitudes 32.5 and 45 against 11 to 14 and 23 to 26 (DETECTOR); the ratio $T(24)/T(12) = 1.677$ against 1.82 to 2.18 (COMPUTATION from two means of unclosed loops' spacings, not a reading of the force's exponent; the two loops are not similar figures, their mean radii 24.3 and 37.6, CONVERSION, so the $1/r$ form's scale symmetry is not tested by them); 4 inside, 7 outside, none moved, 0 record checks failed; the registered D3 row at $n = 9$ (7 inside, 9 outside; Section 4.2's 1.997) stays as history beside it	not yet read against nature: the law's number 1.677 under the one constant and 1.512 under the line drive (COMPUTATION on the host's tick of unclosed loops; the register's 2.00 ± 0.18 a host-tick period; the D3 probes left a confining push and no loop closed, the face clicks DETECTOR); the law as built 1.997 in [1.82, 2.18] on the plane's $1/r$ form (Section 4.2), the inverse square in space NOT MADE; read under the declaration of the weight rule (the limitations paragraph). The run of 2026-09-23 [10]: the click algebra's rung confirmed by the control (979 then 980, DETECTOR); the conditional pins, the law's shell-mean form on an input no click read, FAIL on this fan by the comb (2.5 to 4 times, DETECTOR against a conditional pin); nature's period not compared

S2 The check of the roads

Table S2: The check of the roads (record 817): one row per formula the paper arrives at, the inputs its chain used, the file and line where the chain starts (the click frame on **main** at 70e9781a [30]; Einstein Outside at e8432e6c [15], whose section III is the same check in the source) and the word. No row lists Einstein’s or Lorentz’s formula among its inputs; “derived” stands in no row.

The formula arrived at	The inputs its chain used	Where the chain starts	The word
$W = E_0'^2 + 3\mathbf{p} \cdot \mathbf{p}$ to second order	(A1), (A2); the walk’s invariant $\cos \omega = \cos m \cos \kappa$; the Planck map on main ; the unit $E' = E/c^2$	the click frame, section 7	shown to second order (the conversion’s identity); declared for the rows’ dynamics (record 270)
the boost, Lorentz’s form	(A1); two legs at one Node per interval; $k_{AB} = k_{BA}$	the click frame, section 2 (b) :278–297; section 0 :84–132 (the group up to scale)	follows; Lorentz’s form recovered, the scale r free; (A2) fixes r at $\sqrt{1-v^2}$
$1/\gamma$, the rate r	Theorems 1, 2; r read from two factors	Einstein Outside, II.1 :647	follows in r ; $\sqrt{1-v^2}$ recovered under the line, shown to second order under (A2), declared under the identity, 1 on the law
$E^2 = E_0^2 + c^2 p^2$	$E = E_0/r$, $p = Ev$, the line of II.1; the frame’s section 7 for W	Einstein Outside, II.6 :887	follows as the line; W shown to second order; exact by declaration under the identity; $E_0 = mc^2$ a declared unit
the Outside step, place apart and count apart of a detector, Einstein’s step its most general form	(A1), (A3), the Inside step, Theorems 1, 2	Einstein Outside, section 3 (a) :481 (Einstein’s form named at :497 as the comparison)	follows; Einstein’s form recovered where r is supplied, never an input
Newton’s step as the limit	Theorem 4, the shell mean, $v \ll c$	Einstein Outside, section 3 (c) :554	recovered (the limit)

S3 Reproduction: the seven confirmations

The seven confirmations. Each a run’s clicks against the formula shown Inside, the rows in Tables S1 and 1: the pace, series Q, 290 of 290 primitive directions ($|a| + |b| + |c| \leq 6$) at the derived interval, Node and face, 0.5718 to 0.5893 at finite ages; the click’s power, series L, the (3,4) split 63/1, 31/1, 125/3 at $N_\phi = 64$, 32, 128, the far pair’s cells 27, 5, 5, 27; Mach-Zehnder, 64/0 equal arms, 0/64 at a half turn, 32/32 at a quarter, over 64 births; the marginals, 32/64 per party in every bin at $N_\phi = 64$; GHZ, the four allowed triples 16 each, the products +1 and –1, the others 0; Malus (A12), 128, 0 and 219 of 256 at 45, 90 and 22.5 degrees; light beside a mass, series K, three worlds, the deflection 0.000 pixel, the delay 0.00 interval [3, 6].

S4 The symbols

The symbols.

Table S3: The symbols: physics’ letters keep physics’ meanings, the law’s grains are N with a subscript, a world key is in code font.

Symbol	Kind	Name
$E, E_0, m, \mathbf{p}, c, v, \beta$	scalars, one vector	the energy, the rest energy $E_0 = E_0' c^2$ with $E_0' = N_l N_w M$ the rest energy in whole units, the mass $m = N_l N_w M$ in label units, the momentum, the limit speed $c = 1/\sqrt{3}$ Links per interval, the speed, v/c

Symbol	Kind	Name
$M, N_l, N_w, N_\phi, N_D, N_\theta, N_u, N_t$	scalars	the content (a count of units), the label's scale, the width, the phase circle, the direction bound, the fan's angular grain, the birth wheel, the cosine tables' scale (the law's documents write Q, S and N for N_l, N_w and N_ϕ until their writers' next pass [5])
$\Phi; s, r, d, a, e$	operator; scalars	the interval's map; one accumulator of the state, its rate, its wall (d_p the drive's), its cap, its event count
$\mathbf{G}, U_s, R(f), R(o_A, o_B)$	a matrix, a matrix written plain, a form, a scalar	the click's Gram matrix, the label rotation (Section 2's matrix, written plain there), the click's reading (weight), a cell's weight
$D, \mathbf{u}_D, T_D, S_1, K$	an integer vector written plain, a vector, scalars	a direction, its unit vector at the scale N_l , its resolution, its Manhattan length, the number of directions of a fan (the ladder's K of C_K and b_K is Section 2's)
z, G, ρ	scalars	the redshift, Newton's constant, a family's charge per unit of content
$\mathbb{Z}_{N_\phi}, \mathbb{Z}[\mathbb{Z}_{N_\phi}], \mathbb{Z}[\zeta_{N_\phi}], \mathbb{Z}^2 \otimes \mathbb{Z}^2, B_3$	a cyclic group, a group ring, a ring, a module, a group	the phase circle, the turn its shift; the arrivals' group ring, the merge its addition; its quotient by $x^{N_\phi/2} + 1$, the cyclotomic integers, where opposite phases cancel; the pair's arms under the settings' integer matrices; the six Ports' hyperoctahedral group, 48 signed permutations, their rotations the group of order 24 (Theorem 1); the flight the translation group's shift (Section 5.1)

S5 Auxiliary proofs

Proof of Theorem 1. A map of the six Ports that keeps opposite Ports opposite is a signed permutation of the three axes, $3!2^3 = 48$ of them, the hyperoctahedral group B_3 ; the determinant is a homomorphism onto $\{+1, -1\}$, so its kernel, the group of order 24, has index 2 and acts faithfully on the cube's four body diagonals, so it is the symmetric group S_4 on them; a pseudoscalar column goes to $\det(g)$ times itself by definition.

Proof of Theorem 4. $\sum_i (wa_i)^2 / (\mathfrak{m}A) = w^2 / \mathfrak{m}$. The conjugate transpose maps the outputs $(wa_i, \mathfrak{m}A, p + t_i)$ to $(Aw, A^2\mathfrak{m}, p) \sim (w, \mathfrak{m}, p)$; for two inputs at one splitter the cross terms carry phases differing by $N_\phi/2$ and cancel in the normal form, which the check with the pairs (20, 21), (3, 4, 5) and (119, 120, 169) confirms integer by integer [2], and the engine's merge reproduces on its own rows [1]. That cancellation is the joint condition $B^H B = 2I$ of the balanced splitter's table $\begin{pmatrix} 1 & x^{N_\phi/4} \\ x^{N_\phi/4} & 1 \end{pmatrix}$, and the condition is needed: two

inputs declared (1, 1) with the turns (0, 0) onto the same two outputs have the table $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$, which sends (1, -1) to 0 and merges the inputs (1, 3) and (2, 2) to the same (4, 4) (the exact witness of the independent read of 2026-09-22); each column keeps its own $w^2/\mathfrak{m}$ there and the joint map does not, and Rule S as a generic rule admits such a table; no world of this paper declares one. For U_s the diagonal of $U_s^T U_s$ is $C'^2 + S'^2$ and the off-diagonal $C' S' - S' C' = 0$ in integers.

Proof of Theorem 5. F maps a basis element of Q to a basis element, the Node moved by an offset that is a function of D and of the age, the history's, so with the history fixed it is injective on the basis; S absorbs the arriving row and re-emits its weight, multiplicity and phase on its outputs at age 0, injective on Q by Theorem 4 under the joint condition $B^H B = AI$; R is injective because the turn is a bijection and $U_s^T U_s = n_s I$; M is the identity on the module; a composition of injective linear maps is injective. The collision, a permutation of the amount-1 rows at a Node as a function of the whole set present, is a bijection of states and not a linear map, which is why it is excluded from the linear statement; the engine's inverse interval is the witness on worlds without a measured event [1].

S6 A hand-worked update, and the code

A hand-worked update. A row on the heading $+x$ has the flight accumulator's rate $2S_1N_l = 128$ ($S_1 = 1$, $N_l = 64$) against the wall $2T_D$ with $T_D = \lfloor \sqrt{3 \cdot 64^2} \rfloor = 110$, so $2T_D = 220$, started at $T_D = 110$. Interval by interval the accumulator reads 238 (one wall counted: a Link, the remainder 18), 146 (none), 274 (a Link, 54), 182 (none), 310 (a Link, 90), 218 (none), 346 (a Link, 126), 254 (a Link, 34): Links at the intervals 1, 3, 5, 7, 8, $\dots$, five Links in eight intervals, and in the long run $128/220 = 0.5818$ Links per interval, the heading's pace of Section 4.1. The same three lines of Eq. (1) run the phase (the rate the family's turn, the wall N_ϕ), a body's owed count and every other component; only the rate and the wall differ.

In the code. The engine realises Eq. (1) as one accumulator per component with its rate and wall read from the world file, the carry as the event, in six steps per interval: departures become arrivals (the walk), the readings at each Node, the collision, the measured events' tables and the detectors, the self-creations, and the border's lifetime with the merge of identical rows [1, 14]. The same engine runs every world of this paper from its world file.

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